

AI TOOLS FOR RESEARCH

Student Instruction Sheet

Create Your Own Awesome Research Repository on GitHub

From an AI-assisted research paper to a verified, curated and reusable research resource

Activity Goal

Learn GitHub as a research platform by creating an individual Awesome-style repository around your assigned research topic. Your repository will organize your earlier AI-assisted paper, citation-integrity audit, verified papers, datasets, tools, implementations, and learning resources.

Course	AI Tools for Research
Activity	GitHub Research Curation and Documentation
Nature	Individual Activity
Submission	Public GitHub Repository URL + GitHub Profile URL

1. Learning Outcomes

- Create and maintain a professional GitHub repository.
- Use Markdown to document and organize research resources.
- Curate scholarly papers, datasets, tools, and implementations by research subtopic.
- Verify research references instead of accepting AI-generated citations blindly.
- Use commits to show the development history of research work.
- Create a reusable research resource that another student or researcher can understand.

2. What Is an “Awesome” Repository?

An Awesome-style repository is a carefully curated and well-organized collection of useful resources around one topic. The main page is usually a structured **README.md** containing categorized links, short descriptions, and navigation.

Example

Instead of creating a repository called “assignment-2”, a student working on Explainable AI may create “awesome-explainable-ai” and organize resources under Survey Papers, Foundational Methods, Medical Applications, Datasets, Tools, Implementations, and Tutorials.

For this assignment, you are creating an **Awesome-style academic repository**. You are not required to submit it to the official Awesome project.

3. Use the Same Topic as Your Earlier Research Activity

Continue with the research topic allocated to you earlier. Your new repository should connect the previous work with this GitHub activity.

- AI-assisted research paper
- Citation-integrity / claim-citation audit
- Verified literature and resources collected for the same topic

4. Step-by-Step Instructions

Step 1: Create or Update Your GitHub Account

1. Go to <https://github.com/> and create an account if you do not already have one.
2. Use a professional username wherever possible.
3. Update your profile with your full name, institution, academic status, and a short research-interest bio.

Professional Username Examples

Good: rahul-sharma-ai, ananya-research, r-kumar-ml
 Avoid: killer007, coolboy123, assignmentuser1

Step 2: Create a Public Repository

4. Click New Repository on GitHub.
5. Create a public repository.
6. Use the naming convention: awesome-topic-name.
7. Add a one-line repository description.

8. Initialize the repository with a README.md file.

Recommended repository names:

- awesome-explainable-ai
- awesome-federated-learning
- awesome-medical-image-analysis
- awesome-multimodal-learning

Step 3: Build the Repository Structure

Use a clear structure similar to the following. You may add folders if your topic requires them.

```
awesome-topic-name/
|-- README.md
|-- paper/
|   |-- AI_Assisted_Research_Paper.pdf
|-- citation-audit/
|   |-- Citation_Integrity_Audit.pdf
|-- references/
|   |-- references.md
|-- datasets/
|   |-- datasets.md
|-- tools/
|   |-- tools.md
|-- implementations/
|   |-- github-repositories.md
`-- LICENSE
```

5. README.md: The Most Important Part of Your Repository

The README is the homepage of your repository. It must allow a visitor to understand the topic and navigate the resource collection without opening every file.

Required Section	What to Include
Repository Title	Example: Awesome Explainable Artificial Intelligence
Short Description	2–4 sentences explaining the scope and purpose of the repository.
Table of Contents	Clickable links to the major sections.
Topic Overview	Approximately 150–300 words describing the area, importance, key problems, and major directions.
AI-Assisted Research Paper	Paper title, short abstract/description, and repository link to your own paper.
Citation Integrity Audit	Short statement explaining that references and claims were checked, with a link to your audit.
Curated Research Papers	Verified papers organized into meaningful categories.

Required Section	What to Include
Datasets	Relevant datasets with source, description, use, and link.
Tools and Libraries	Useful software/frameworks with brief descriptions.
GitHub Implementations	Existing high-quality implementations relevant to your topic.
Tutorials and Learning Resources	Authoritative learning material, documentation, lectures, or benchmark resources.
License	Appropriate license information for your own repository content.

6. Research Paper Collection

Your repository must contain at least **20 verified scholarly papers**. More papers do not automatically mean better work. Quality, relevance, verification, and organization are more important than quantity.

Suggested Categories

- Survey and Review Papers
- Foundational Papers
- Recent Research Papers
- Methods / Algorithms
- Applications
- Evaluation Methods / Benchmarks

Required Format for Each Paper

- **Paper Title**
Authors, Year, Journal/Conference
[Paper / DOI](link)
One-line explanation of why the paper is relevant.

Do Not Paste Only URLs

Every listed paper must have enough information for another researcher to understand what it is and why it belongs in your collection.

7. Citation and Resource Verification

Before adding a scholarly paper, verify the following:

- Correct title
- Correct authors
- Publication year
- Journal or conference
- DOI, where available
- Whether the paper genuinely exists

- Whether the provided link corresponds to the same paper

Recommended verification sources include publisher websites, DOI/Crossref, Google Scholar, Semantic Scholar, PubMed (where relevant), and other authoritative scholarly databases.

Core Rule

Never accept a reference merely because an AI tool generated it. AI may help discover a resource, but you are responsible for verifying it.

8. Minimum Supporting Resources

Resource Type	Minimum	Expected Information
Scholarly papers	20	Verified, categorized, linked, and briefly explained
Datasets	3	Name, source, description, application, link
Tools / libraries	5	Purpose and official/project link
GitHub implementations	5	What it implements and why it is relevant
Tutorial / learning resources	5	Authoritative resource with brief purpose
Meaningful commits	5	Commits that show genuine development

If datasets are genuinely not applicable to your topic, state this clearly in the repository and explain why.

9. How to Select Existing GitHub Repositories

Do not choose repositories only because they appear first in a search or have many stars. Examine:

- Documentation quality
- Availability and clarity of source code
- Recent maintenance/activity
- Examples or tutorials
- Reproducibility
- License
- Connection to a research paper or recognized project

10. Copyright and Ethical Use

Do not upload copyrighted PDFs of papers written by other authors unless redistribution is clearly permitted. Prefer linking to the DOI, publisher page, PubMed record, arXiv page, or official open-access source.

You may upload your own assignment paper and your own citation-audit document.

11. Learn GitHub, Not Just File Uploading

Your repository should show development over time. Make at least five meaningful commits.

Good Commit Messages

- Initial repository structure
- Add foundational research papers
- Add verified datasets
- Update citation audit section
- Add GitHub implementations
- Improve README documentation

Avoid

- abc
- done
- final
- changes
- update1

12. Basic Markdown You Should Know

Purpose	Markdown
Heading	# Main Heading ## Section ### Subsection
Bold	**Important**
Italics	<i>*Research*</i>
Link	[Semantic Scholar](https://www.semanticscholar.org/)
Bullets	- Resource 1 - Resource 2
Numbered List	1. Step one 2. Step two

13. Suggested README Template

```
# Awesome [Research Topic]
```

```
A curated collection of research papers, datasets, tools,
implementations and learning resources related to [topic].
```

```
## Contents
- Overview
- AI-Assisted Research Paper
- Survey Papers
- Foundational Papers
- Recent Research
- Datasets
- Tools and Libraries
- GitHub Implementations
- Tutorials
- Citation Integrity Audit
```

```
## Overview
Brief introduction...
```

```
## AI-Assisted Research Paper
```

```
[View Paper](paper/AI_Assisted_Research_Paper.pdf)

## Survey Papers
- ...

## Foundational Papers
- ...

## Recent Research Papers
- ...

## Datasets
- ...

## Tools and Libraries
- ...

## GitHub Implementations
- ...

## Tutorials
- ...

## Citation Integrity Audit
[View Audit](citation-audit/Citation_Integrity_Audit.pdf)

## License
...
```

14. Use of Generative AI

You may use generative AI to help with:

- Understanding GitHub concepts
- Learning Markdown
- Suggesting an initial repository structure
- Brainstorming categories
- Improving descriptions
- Finding candidate resources
- Troubleshooting formatting

However, the final repository must demonstrate **human verification and curation**. Simply asking an AI system to create an Awesome repository and pasting the output is not sufficient.

15. Final Submission Checklist

- ☐ Professional GitHub profile completed
- ☐ Public repository created with correct name
- ☐ Repository description added
- ☐ README.md complete
- ☐ Clickable Table of Contents

- ☐ Topic overview included
- ☐ AI-assisted research paper linked
- ☐ Citation-integrity audit linked
- ☐ At least 20 verified scholarly papers
- ☐ Papers categorized meaningfully
- ☐ One-line description for each paper
- ☐ At least 3 datasets, where applicable
- ☐ At least 5 tools/resources
- ☐ At least 5 GitHub implementations
- ☐ At least 5 learning resources
- ☐ No fabricated references
- ☐ No unauthorized copyrighted PDFs
- ☐ At least 5 meaningful commits
- ☐ No obviously broken links
- ☐ Repository remains public for evaluation

16. What to Submit

Submit the following through the course submission mechanism:

1. Student Name
2. Roll Number
3. Assigned Research Topic
4. GitHub Profile URL
5. Awesome Repository URL
6. Repository Name

Example Repository URL

<https://github.com/username/awesome-explainable-ai>

17. Evaluation Rubric

Component	Marks
GitHub profile and repository setup	5
Repository organization	10
README quality and Markdown	15
Research-paper collection	15
Categorization and curation	15

Component	Marks
Verification and citation integrity	15
Datasets, tools and implementations	10
Quality of descriptions	5
Git/commit practices	5
Overall professionalism and usability	5
TOTAL	100

18. Important Rules

Rule 1: Do not fabricate references or resources.

Rule 2: Do not copy another student's repository.

Rule 3: Do not blindly paste AI-generated lists.

Rule 4: Verify scholarly resources independently.

Rule 5: Do not upload copyrighted research-paper PDFs without permission.

Rule 6: Keep the repository public and accessible for evaluation.

Rule 7: Commit history should reflect genuine development.

Rule 8: The repository should be understandable to someone who has not attended the course.

19. Final Goal

By the end of this activity, you should be able to say:

"I can use GitHub to build, document, curate, verify, maintain, and share a structured research knowledge resource."